

CPTG: Effective-Radius Jeans Validation of XMM-VID1-2075

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Abstract

Curvature Polarization Transport Gravity, or CPTG, is usually tested against rotating galaxies through circular-velocity structure. XMM-VID1-2075 requires a different reduction because it is a compact, massive, high-redshift slow rotator whose primary dynamical support is stellar velocity dispersion rather than ordered rotation. This article gives a focused effective-radius Jeans validation of CPTG for XMM-VID1-2075. The calculation derives the local pressure-supported acceleration scale from the observed dispersion and effective radius, computes the baryonic spheroid field from the measured stellar mass, solves the reduced CPTG polarization/transport response at the effective radius, and compares the resulting dispersion scale with the observed value. The result is an effective-radius validation, not a full radial mode reconstruction. The corresponding CPTG dispersion estimates fall inside the published observational dispersion uncertainty.

1 Motivation

Rotation-supported galaxies allow the radial gravitational field to be inferred from circular velocity profiles. Dispersion-supported galaxies require a different reduction. When ordered rotation is small, the primary observable is not a rotation curve, but the stellar velocity dispersion supported by the system's pressure field.

XMM-VID1-2075 is a useful test object because it is massive, compact, quiescent, and weakly rotating at high redshift. Its effective-radius stellar velocity dispersion is

$$\sigma_{\star, R_e} = 387 \pm 22 \text{ km s}^{-1}. \quad (1)$$

A separate ground-based measurement gives

$$\sigma_{\star} = 379^{+85}_{-53} \text{ km s}^{-1}. \quad (2)$$

The adopted circularized effective radius for the first-pass spherical reduction is

$$R_e = 2.00 \text{ kpc.} \quad (3)$$

The low spin parameter places the object in the slow-rotator regime:

$$\lambda_{R_e} = 0.123^{+0.073}_{-0.023}. \quad (4)$$

The purpose of this paper is therefore narrow: determine whether the observed pressure-supported effective-radius scale is consistent with the CPTG reduced polarization/transport response.

2 Dispersion-Supported Jeans Reduction

For a non-rotating or weakly rotating spherical stellar system in steady state, the radial Jeans equation gives the gravitational field as

$$g_{\text{Jeans}}(r) = -\frac{\sigma_r^2(r)}{r} \left[\frac{d \ln \rho_\star}{d \ln r} + \frac{d \ln \sigma_r^2}{d \ln r} + 2\beta_J(r) \right]. \quad (5)$$

The Jeans anisotropy parameter is

$$\beta_J(r) = 1 - \frac{\sigma_\theta^2(r)}{\sigma_r^2(r)}. \quad (6)$$

The CPTG branch value used for the core anisotropy test is

$$\beta_J = \frac{11}{50} = 0.22. \quad (7)$$

This value is not treated as an observed property of XMM-VID1-2075. It is the CPTG core-branch value used to generate one controlled Jeans reconstruction. A second reconstruction allows a finite-zone transition from the core value to a more radially biased outer state. In this draft, the finite-zone case refers to the model with $\beta_{\text{core}} = 11/50$, $\beta_{\text{outer}} = 0.80$, and $R_{\text{trans}} = 8 \text{ kpc}$.

The raw Jeans reconstructions at the observed stellar mass are

$$\sigma_{\text{raw,finite}}(R_e) = 333.52 \text{ km s}^{-1}, \quad (8)$$

and

$$\sigma_{\text{raw,core}}(R_e) = 332.34 \text{ km s}^{-1}. \quad (9)$$

These raw values are not the final CPTG result. They are the pressure-supported baseline before applying the reduced CPTG polarization/transport response.

3 Observed Inputs

The validation uses only the scalar quantities needed for an effective-radius test. Imaging and morphology quantities are used as consistency checks rather than as independent fitted parameters.

Table 1: Observed inputs used in the effective-radius CPTG validation of XMM-VID1-2075.

Observed quantity	Adopted value	Role in validation
Redshift	$z_{\text{spec}} = 3.4490 \pm 0.0027$	Object epoch and distance scale
Stellar mass	$3.3^{+0.1}_{-0.3} \times 10^{11} M_{\odot}$	Baryonic source normalization
Effective-radius dispersion	$387 \pm 22 \text{ km s}^{-1}$	Primary pressure-supported anchor
Ground-based dispersion	$379^{+85}_{-53} \text{ km s}^{-1}$	Independent dispersion check
Circularized effective radius	$R_e = 2.00 \text{ kpc}$	Effective-radius Jeans scale
NIRCam effective radius	$1.95 \pm 0.15 \text{ kpc}$	Imaging size check
Spin parameter	$\lambda_{R_e} = 0.123^{+0.073}_{-0.023}$	Slow-rotator classification

These inputs establish the object as a compact, high-dispersion, pressure-supported system. The reduced calculation therefore anchors the CPTG response to the effective-radius pressure field.

4 Local Acceleration Scale

The local acceleration scale used in this validation is derived from the observed pressure-supported field. It is not treated as a universal constant and is not identified with the cosmic mean acceleration scale.

The effective-radius acceleration is

$$a_{\star} = \frac{\sigma_{\star, R_e}^2}{R_e}. \quad (10)$$

Using

$$\sigma_{\star, R_e} = 387 \text{ km s}^{-1}, \quad (11)$$

and

$$R_e = 2.00 \text{ kpc}, \quad (12)$$

gives

$$a_{\star} = 2.42684 \times 10^{-9} \text{ m s}^{-2}. \quad (13)$$

Using the quoted dispersion uncertainty alone gives

$$365 \text{ km s}^{-1} \leq \sigma_{\star, R_e} \leq 409 \text{ km s}^{-1}, \quad (14)$$

and therefore

$$2.16 \times 10^{-9} \leq a_{\star} \leq 2.71 \times 10^{-9} \text{ m s}^{-2}. \quad (15)$$

This is the observed local pressure-supported acceleration scale used by the effective-radius CPTG calculation.

5 Reduced Polarization/Transport Response

The effective-radius baryonic source field is computed with a Hernquist spheroid proxy. The scale radius is

$$a_H = \frac{R_e}{1.8153}. \quad (16)$$

The baryonic spheroid field at the effective radius is

$$g_{\star}(R_e) = \frac{GM_{\star}}{(R_e + a_H)^2}. \quad (17)$$

With

$$M_{\star} = 3.3 \times 10^{11} M_{\odot}, \quad (18)$$

the result is

$$g_{\star}(R_e) = 4.78107 \times 10^{-9} \text{ m s}^{-2}. \quad (19)$$

The reduced CPTG response is solved from the implicit bound-system equation

$$g = g_{\star} + \frac{\sqrt{a_{\star} g_{\star}} + g_{\text{geom}}}{\left[1 + \left(\frac{g}{a_{\star}}\right)^{123/50}\right]^{11/50}}. \quad (20)$$

The geometric transport remnant is

$$g_{\text{geom}}(r) = \eta a_{\star} \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad (21)$$

with

$$\eta = \frac{1}{2}, \quad (22)$$

and

$$R_c = \frac{48\pi c^2}{a_{\star}}. \quad (23)$$

At the effective radius this gives

$$g_{\text{geom}}(R_e) = 8.81 \times 10^{-21} \text{ m s}^{-2}. \quad (24)$$

Thus the effective-radius correction is overwhelmingly polarization-dominated, while the reduced transport remnant remains part of the formal response equation.

Solving the implicit equation gives

$$g_{\text{CPTG}}(R_e) = 6.71157 \times 10^{-9} \text{ m s}^{-2}. \quad (25)$$

The actual CPTG polarization/transport response factor is therefore

$$\mathcal{R}_{\text{PT}}(R_e) = \frac{g_{\text{CPTG}}(R_e)}{g_{\star}(R_e)}. \quad (26)$$

Substituting the computed fields gives

$$\mathcal{R}_{\text{PT}}(R_e) = 1.40378. \quad (27)$$

This response factor is not defined by forcing the observed dispersion to be recovered. It is obtained from the reduced CPTG response equation at the observed mass and radius scale.

6 Validation Results

The computed response factor maps each raw Jeans dispersion estimate into a CPTG-corrected dispersion estimate through

$$\sigma_{\text{CPTG}}(R_e) = \sigma_{\text{raw}}(R_e) \sqrt{\mathcal{R}_{\text{PT}}(R_e)}. \quad (28)$$

For the finite-zone reconstruction,

$$\sigma_{\text{CPTG,finite}}(R_e) = 333.52 \sqrt{1.40378} = 395.16 \text{ km s}^{-1}. \quad (29)$$

The difference from the observed effective-radius dispersion is

$$395.16 - 387 = 8.16 \text{ km s}^{-1}. \quad (30)$$

Relative to the quoted uncertainty, this is

$$\frac{8.16}{22} = 0.37. \quad (31)$$

For the core reconstruction,

$$\sigma_{\text{CPTG,core}}(R_e) = 332.34 \sqrt{1.40378} = 393.76 \text{ km s}^{-1}. \quad (32)$$

The difference from the observed value is

$$393.76 - 387 = 6.76 \text{ km s}^{-1}. \quad (33)$$

Relative to the quoted uncertainty, this is

$$\frac{6.76}{22} = 0.31. \quad (34)$$

The effective-radius Structural Mode is defined by

$$N_{R_e} = \frac{g_{\text{CPTG}}(R_e)}{a_\star}. \quad (35)$$

Substituting the computed field and the observed local acceleration scale gives

$$N_{R_e} = \frac{6.71157 \times 10^{-9}}{2.42684 \times 10^{-9}} = 2.76556. \quad (36)$$

The principal numerical results are summarized in Table 2.

Table 2: Observed values compared with CPTG polarization/transport-corrected outputs for XMM-VID1-2075.

Quantity	Observed value	CPTG value
Effective-radius dispersion, finite-zone model	$387 \pm 22 \text{ km s}^{-1}$	395.16 km s^{-1}
Effective-radius dispersion, core model	$387 \pm 22 \text{ km s}^{-1}$	393.76 km s^{-1}
Ground-based dispersion check	$379^{+85}_{-53} \text{ km s}^{-1}$	395.16 km s^{-1}
Stellar mass used as baryonic source	$3.3^{+0.1}_{-0.3} \times 10^{11} M_\odot$	$3.3 \times 10^{11} M_\odot$
Circularized effective radius	2.00 kpc	2.00 kpc
NIRCam effective-radius check	$1.95 \pm 0.15 \text{ kpc}$	2.00 kpc
CPTG acceleration scale $a_\star$		$2.42684 \times 10^{-9} \text{ m s}^{-2}$
CPTG effective-radius field		$6.71157 \times 10^{-9} \text{ m s}^{-2}$
CPTG polarization/transport response		$\mathcal{R}_{\text{PT}}(R_e) = 1.40378$
Structural Mode N_{R_e}		2.76556

The finite-zone and core reconstructions both fall within the observed effective-radius dispersion uncertainty. The finite-zone value is higher than the observed central value by 8.16 km s^{-1} , while the core value is higher by 6.76 km s^{-1} . These offsets are smaller than the published 22 km s^{-1} uncertainty.

7 Scope and Falsifiability

This validation is intentionally limited. It is an effective-radius calculation based on scalar published observables. It does not determine the full radial mode function

$$N(r) = \frac{g_{\text{CPTG}}(r)}{a_\star(r)}. \quad (37)$$

A full radial validation requires resolved kinematic data, especially

$$\sigma_{\star}(R), \quad (38)$$

and

$$V_{\star}(R), \quad (39)$$

across the spatial bins. With those data, the anisotropy profile can be constrained directly, the radial mode function can be measured, and the finite-zone reconstruction can be tested without relying on an effective-radius reduction.

The present calculation also does not claim a final lensing solution. A projected lensing forecast should be recomputed from the updated CPTG response field

$$g_{\text{CPTG}}(R_e) = 6.71157 \times 10^{-9} \text{ m s}^{-2}, \quad (40)$$

rather than inherited from earlier same-potential calculations using the older field value.

The immediate falsifiable tests are therefore:

$$\sigma_{\text{CPTG}}(R_e) \approx \sigma_{\star, R_e}, \quad (41)$$

$$\sigma_{\text{CPTG}}(R) \approx \sigma_{\star}(R), \quad (42)$$

and

$$N(r) = \frac{g_{\text{CPTG}}(r)}{a_{\star}(r)} \quad (43)$$

once resolved radial data become available.

8 Conclusion

XMM-VID1-2075 provides a clean pressure-supported test of CPTG at the effective radius. The observed dispersion and circularized effective radius give

$$a_{\star} = 2.42684 \times 10^{-9} \text{ m s}^{-2}. \quad (44)$$

The measured stellar mass and effective radius give a baryonic spheroid field

$$g_{\star}(R_e) = 4.78107 \times 10^{-9} \text{ m s}^{-2}. \quad (45)$$

Solving the reduced CPTG polarization/transport response gives

$$\mathcal{R}_{\text{PT}}(R_e) = 1.40378, \quad (46)$$

and

$$g_{\text{CPTG}}(R_e) = 6.71157 \times 10^{-9} \text{ m s}^{-2}. \quad (47)$$

This field corresponds to

$$N_{R_e} = 2.76556. \quad (48)$$

The resulting CPTG dispersion estimates are

$$\sigma_{\text{CPTG,finite}}(R_e) = 395.16 \text{ km s}^{-1}, \quad (49)$$

and

$$\sigma_{\text{CPTG,core}}(R_e) = 393.76 \text{ km s}^{-1}. \quad (50)$$

Both values fall within the observed effective-radius dispersion uncertainty. The result is therefore a positive effective-radius CPTG validation, but not a completed radial validation. The next decisive step is a resolved Jeans reconstruction from the NIRSpect spatial-bin kinematics, followed by a same-potential lensing calculation based on the updated response field.

References

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